Lubuntu 16.04 installation guide for Vortex86DX3

(2017/07/26)

Preparation of Installation

1. Host computer (Standard x86 PC)
2. Storage device (Recommended at least 8GB)
3. Oracle VM VirtualBox
   Please download it from VirtualBox website.
   <https://www.virtualbox.org/>
4. Lubuntu 16.04 install image
   Please download 'lubuntu-16.04.1-desktop-i386.iso' from Lubuntu website.
   <http://cdimage.ubuntu.com/lubuntu/releases/16.04/release/>
5. Customized Linux kernel package
   Please download 'custom-linux-kernel-package-lubuntu-16.04-20170726.iso' from FTP site.
   <ftp://ftp.dmp.com.tw/Demo/Linux/Lubuntu-16.04/custom-linux-kernel-package-lubuntu-16.04-20170726.iso>

Installation procedures

This section describes all the procedures to install Lubuntu 16.04 on a virtual machine and configure it for the Vortex86DX3 platform. It includes installing virtual machine software, creating a new virtual machine, installing Lubuntu on the virtual machine, and customizing the Lubuntu system.

Installing VirtualBox

Please install VirtualBox on Host computer. Please also refer to VirtualBox manual for details.

Creating a new virtual machine

Please create a virtual machine in VirtualBox. The recommended settings are below.

- Type: Linux
- Version: Ubuntu (32-bit)
- Memory: 1024 MB
- Virtual Hard Disk: VDI, Dynamically allocated, 7.00GB

Installing Lubuntu 16.04 on virtual machine

1. Attach 'lubuntu-16.04.1-desktop-i386.iso' to the virtual machine.
2. Start the virtual machine. It will boot from Lubuntu installation image. Then choose 'Install Lubuntu' in welcome screen menu.
3. Follow the steps on screen to install a standard Lubuntu 16.04 system.
4. When installation complete, the Lubuntu installer will prompt to restart the virtual machine. Please restart the virtual machine to boot into the new installed Lubuntu system.

Modifying Lubuntu system settings

1. Add the following line into '/etc/systemd/logind.conf'.

   ```
   HandleLidSwitch=ignore
   ```

2. Edit '/etc/default/grub'. Add 'pci=nomsi' to the GRUB_CMDLINE_LINUX variable.

   ```
   GRUB_CMDLINE_LINUX="pci=nomsi"
   ```

3. Create a new '99-screen.conf' file with the following content in the '/usr/share/X11/xorg.conf.d/' directory.

   ```
   Section "Device"
     Identifier  "Device0"
     Driver      "vesa"
   EndSection

   Section "Monitor"
     Identifier  "Monitor0"
   EndSection

   Section "Screen"
     Identifier  "Screen0"
     Monitor     "Monitor0"
     Device      "Device0"
   EndSection
   ```

4. Add the following line into '/etc/sysctl.d/99-sysctl.conf'.

   ```
   vm.swappiness=1
   ```

Installing customized Linux kernel package and extra package

1. Attach 'custom-linux-kernel-package-lubuntu-16.04-20170726.iso' to the virtual machine.
2. Mount disc.
3. Install customized linux-headers and linux-image packages in the mount point of the disc.

   ```
   $ sudo dpkg -i /mount_point/linux-headers-4.4*.deb
   $ sudo dpkg -i /mount_point/linux-image-4.4.6_4.4.6-1_i386.deb
   ```

   (Please replace the 'mount_point' with the correct one.)

4. Install the 'dkms' package.

```
$ sudo apt-get update
$ sudo apt-get install dkms
```

5. Install kernel modules.

```
$ sudo dpkg -i /mount_point/alim6117-wdt-dkms_0.3.0_all.deb
$ sudo dpkg -i /mount_point/vdx3-temp-dkms_1.0.0_all.deb
```

6. Download and install the 'fake-pae' package.

```
$ wget ftp://ftp.dmp.com.tw/Demo/Linux/Lubuntu-16.04/fake-pae_1.0.2_i386.deb
$ sudo dpkg -i fake-pae_1.0.2_i386.deb
```

7. After installed all packages, please shutdown the Lubuntu system.

Post-installation

The section shows a way to convert the virtual disk file of VirtualBox to a raw disk image file and write the image file to a storage device on Host computer.

Converting the virtual disk image on Host OS

1. Find the file path of 'VBoxManage'. Under Windows, it is located in the same directory as the VirtualBox installation directory.
2. Find the file path of the virtual machine disk. Under Windows, please right click on the Lubuntu 16.04 virtual machine item, and choose 'Show in Explorer'.
3. Open a command prompt or a terminal.
4. Run the following command to make the raw disk image of the customized Lubuntu system.

   ```
   VBoxManage clonehd "lubuntu1604.vdi" "lubuntu1604.raw" --format raw
   ```

   ('lubuntu1604.vdi' is only an example filename, please replace it with the valid filename.)

Writing the raw disk image file

Please use a dd-like tool or any disk image writing program (win32diskimager or Rufus) to write the image file to a storage device.

Building custom Linux kernel packages for Vortex86DX3

(2017/07/26)

Prerequisites

Lubuntu 16.04 (i386) on VirtualBox

At least 20GB of free storage space

Setting up a kernel build environment

1. Booting with a standard Lubuntu kernel (e.g. Linux 4.4.0-xx-generic).
2. Adding source package entries into "/etc/apt/sources.list".
3. Installing required programs.

   ```
   $ sudo apt-get install build-essential libncurses5-dev
   ```

4. Running the command below to get all building related packages ready.

   ```
   $ sudo apt-get build-dep linux-image-$(uname -r)
   ```

5. Running the command below to get the source code of the Ubuntu kernel.

   ```
   $ apt-get source linux-image-$(uname -r)
   ```

Building the kernel packages

Downloading the kernel configuration file for Vortex86DX3 platform. Then copying the file to Linux kernel directory.

```
$ wget ftp://ftp.dmp.com.tw/Demo/Linux/Lubuntu-16.04/config-4.4.6-vortex86dx3.gz
$ export KERNEL_REL=linux-$(uname -r | cut -f 1 -d '-')
$ zcat config-4.4.6-vortex86dx3.gz > ${KERNEL_REL}/.config
```

Compiling and building the kernel packages.

```
$ cd ${KERNEL_REL}
$ make olddefconfig
$ make deb-pkg LOCALVERSION=-vortex86dx3
```

When finished, the kernel package files should be located in '../*.deb'.

Setting up a kernel module build environment

1. Setting environment variables.

```
$ export KERNELRELEASE=$(cat < include/config/kernel.release)
```

2. Installing a kernel headers package.

```
$ cd ..
$ dpkg -i linux-headers-4.4.*.deb
```

3. Installing the 'dkms' package.

```
$ sudo apt-get install dkms
```

Building kernel module packages

Watchdog driver

Downloading a watchdog driver source file.

```
$ wget ftp://ftp.dmp.com.tw/Demo/Linux/Lubuntu-16.04/alim6117_wdt-0.3.0-
dkms.tar.gz
```

Unpack the source package and copy the files into "/usr/src/".

```
$ tar -xf alim6117_wdt-0.3.0-dkms.tar.gz
$ sudo cp -a alim6117_wdt-0.3.0 /usr/src
```

Using 'dkms' to build kernel module and debian package.

```
$ sudo dkms add -m alim6117_wdt -v 0.3.0
$ sudo dkms build -m alim6117_wdt -v 0.3.0 -k ${KERNELRELEASE}
$ sudo dkms mkdeb -m alim6117_wdt -v 0.3.0 -k ${KERNELRELEASE}
```

If there is no error, the package file should be located in "/var/lib/dkms/alim6117_wdt/0.3.0/deb/alim6117-wdt-dkms_0.3.0_all.deb".

SoC Temperature driver

Downloading the hwmon driver source file.

```
$ wget ftp://ftp.dmp.com.tw/Demo/Linux/Lubuntu-16.04/vdx3-temp-1.0.0-dkms.tar.gz
```

Unpack the source package and copy the files into "/usr/src/".

```
$ tar -xf vdx3-temp-1.0.0-dkms.tar.gz
$ sudo cp -a vdx3-temp-1.0.0 /usr/src
```

Using 'dkms' to build kernel module and debian package.

```
$ sudo dkms add -m vdx3-temp -v 1.0.0
$ sudo dkms build -m vdx3-temp -v 1.0.0 -k ${KERNELRELEASE}
$ sudo dkms mkdeb -m vdx3-temp -v 1.0.0 -k ${KERNELRELEASE}
```

If there is no error, the package file should be located in "/var/lib/dkms/vdx3-temp/1.0.0/deb/vdx3-temp-dkms_1.0.0_all.deb".

Vortex86DX3 Linux driver guide

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Watchdog

The watchdog timer activates when '/dev/watchdog' is opened; The watchdog timer deactivates when you send a character 'V' to '/dev/watchdog' before it is closed.
The default timeout is set to 60 secs. If you want to change the timeout setting, please specify 'timeout' argument when you load the 'alim6117_wdt' kernel module.
Here are examples:
Load the watchdog module and set the timeout to 120 secs

```
$ sudo modprobe alim6117_wdt timeout=120
```

Activate watchdog

```
$ sudo sh -c 'echo > /dev/watchdog'
```

Ping watchdog (keep watchdog alive)

```
$ sudo sh -c 'echo > /dev/watchdog'
```

Deactivate watchdog

```
$ sudo sh -c 'echo V > /dev/watchdog'
```

SoC Temperature

The SoC temperature can be read from '/sys/devices/platform/vdx3-temp/nbtemp_input'.
The unit of the value is millidegree Celsius.
Here are examples:
Load the vdx3-temp module

```
$ sudo modprobe vdx3-temp
```

Read the temperature value

```
$ cat /sys/devices/platform/vdx3-temp/nbtemp_input
55844
```

In this example, the SoC temperature is 55.844 °C.